```
  j ← i * 2 + 1
  if i % 2 == 0 then
    A[k] ← 1
    A[j] ← 0
  else
    A[k] ← 0
    A[j] ← 1
A[2*n-1] ← 0
```

The `for` loop executes `n` times the 8 lines below it (all but the very last line in the listing above) and each branch of the if statement has 2 instructions (that can be recognized because they are further indented).

The line containing `A[2*n-1] ← 0` does not have any indentation (is at the same level of indentation of the first line) and that shows it is the next instruction to be executed *after* the for loop ends.

A.6 Functions

To group and reuse code, we will use functions.

A function defines a block of code where local variables are in scope. It also has a signature declaring its name and what are the arguments it expects: the variables that are the input to the function. Finally, a function also returns a value, that is effectively the function's output.

Breaking code into functions allows writing reusable code that can be more easily understood and unit-tested,